

AP-E2: Exact Discriminant–Contact Regression First-Principles Projective Charts, Killing Normalization, and the Fail-Closed Route-E Convention Boundary

Route F research note

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Abstract

This note completes AP-E2 as an exact, non-promoting regression theorem. On the conditional carrier $H^0(\mathbb{CP}^1, T_{\mathbb{CP}^1}) \simeq H^0(\mathbb{CP}^1, \mathcal{O}(2)) \simeq \mathfrak{sl}_2(\mathbb{C})$, let $p(u, v) = au^2 + buv + cv^2$ and $A_p = \begin{pmatrix} -b/2 & -c \\ a & b/2 \end{pmatrix}$. The projective convention is fixed, rather than guessed, by the identity $\det(z, A_p z) = p(z)$. Direct multiplication and the adjoint-trace normalization then give

$$A_p^2 = \frac{\Delta(p)}{4} I, \quad B_{\text{Kill}}(A_p, A_p) = 2\Delta(p), \quad (p, p)_2 = -2\Delta(p),$$

where $\Delta = b^2 - 4ac$. Thus, for a nonzero section, two distinct projective zeros, a regular semisimple generator, nonzero Killing norm, and nonzero Route-E contact are equivalent. The boundary $\Delta = 0$ is the nonzero nilpotent double-zero cone; the zero section is a separate excluded stratum.

The note also resolves the convention-sensitive factor of two. In Route-E spherical coordinates $\mathbf{x} = (x_+, x_0, x_-)^T$,

$$B = 2\sqrt{3} \mathbf{x}^T K_{\text{tr}} \mathbf{x},$$

whereas in normalized polynomial coefficients $\mathbf{y} = (a, b/\sqrt{2}, c)^T$,

$$B = 4\sqrt{3} \mathbf{y}^T K_{\text{tr}} \mathbf{y}.$$

These formulas are compatible because $\mathbf{y} = R\mathbf{x}/\sqrt{2}$. A deterministic 30-check audit tests exact rational identities, projective charts, $SL(2)$ covariance, the degenerate boundary, Route-E source hashes, 100-digit complex samples, and four deliberately wrong conventions. All checks pass, but `physics.promotion_allowed=false`: AP-E2 preserves an algebraic theorem and does not derive $H3^+$, a physical two-center interpretation, or the AP-E3 level $k = 2$.

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1 Scope, assumptions, and final logical status

AP-E2 has one narrow purpose: freeze the exact discriminant/contact theorem against later changes of chart, basis, or normalization. It does not add a new action and does not infer dynamics from an invariant polynomial.

The only geometric input is the conditional Route-E branch

$$\Sigma \simeq \mathbb{CP}^1, \quad T_{\mathbb{CP}^1} \simeq \mathcal{O}(2), \quad V := H^0(\mathbb{CP}^1, T_{\mathbb{CP}^1}) \simeq \mathfrak{sl}_2(\mathbb{C}). \quad (1.1)$$

The word “conditional” is essential: the algebra below is unconditional after Eq. (1.1) is assumed, but AP-E2 does not derive the microscopic origin of that carrier or of the stronger contact hypothesis $H3^+$.

We distinguish four statuses.

Status	Meaning
exact theorem	An identity proved from the displayed definitions over $\mathbb{C}$, with every normalization fixed.
regression pass	A deterministic machine check of a theorem, boundary case, source hash, or known convention failure.
conditional carrier	A conclusion valid on Eq. (1.1); it is not a derivation of that equation.
not promoted	A geometric resemblance has not been turned into a physical state-space, action, spectrum, or family-selection mechanism.

Promotion gate 1.1 (AP-E2 non-promotion). *A green AP-E2 regression may certify the exact invariant identity and its regular/nilpotent stratification. It may not identify the two zeros with two particles, opposite energies, or two families; it may not promote $H3$ to $H3^+$; and it may not supply the level-two prequantum bundle required by AP-E3. The output Boolean must therefore remain `physics_promotion_allowed=false`.*

2 First principles on $\mathbb{CP}^1$

2.1 Two charts force a quadratic vector field

Let $[u : v]$ be homogeneous coordinates on $\mathbb{CP}^1$. On $U_u = \{u \neq 0\}$, set

$$\xi = \frac{v}{u}, \quad (2.1)$$

and write a holomorphic vector field as

$$X = q(\xi)\partial_\xi. \quad (2.2)$$

On $U_v = \{v \neq 0\}$, set $\eta = u/v = 1/\xi$. Since $\partial_\xi = -\eta^2\partial_\eta$, the same field is

$$X = -\eta^2 q(1/\eta)\partial_\eta. \quad (2.3)$$

Regularity at $\eta = 0$ forbids powers of ξ above two. Therefore

$$q(\xi) = a + b\xi + c\xi^2, \quad a, b, c \in \mathbb{C}. \quad (2.4)$$

Conversely, Eq. (2.3) becomes

$$X = -(a\eta^2 + b\eta + c)\partial_\eta, \quad (2.5)$$

which is regular. Hence Eq. (2.4) lists every global holomorphic vector field and proves directly that the section space has complex dimension three.

The basis

$$E = \partial_\xi, \quad H = -2\xi\partial_\xi, \quad F = -\xi^2\partial_\xi \quad (2.6)$$

obeys

$$[H, E] = 2E, \quad [H, F] = -2F, \quad [E, F] = H. \quad (2.7)$$

This constructs the isomorphism $H^0(\mathbb{CP}^1, T_{\mathbb{CP}^1}) \simeq \mathfrak{sl}_2(\mathbb{C})$ without dimension matching alone.

2.2 Homogenization and the projective matrix

Homogenizing Eq. (2.4) gives

$$p(u, v) = au^2 + buv + cv^2, \quad p(1, \xi) = q(\xi). \quad (2.8)$$

For $z = (u, v)^T$, define

$$A_p = \begin{pmatrix} -b/2 & -c \\ a & b/2 \end{pmatrix} \in \mathfrak{sl}_2(\mathbb{C}). \quad (2.9)$$

This sign convention is fixed by the following identity.

Lemma 2.1 (Section–eigenline identity). *For every $z = (u, v)^T$,*

$$\boxed{\det(z, A_p z) = p(u, v)}. \quad (2.10)$$

Consequently, $[u : v]$ is a zero of p if and only if it is an eigenline of A_p .

Proof. Equation (2.9) gives

$$A_p z = \begin{pmatrix} -bu/2 - cv \\ au + bv/2 \end{pmatrix}. \quad (2.11)$$

Taking the determinant of the matrix with columns $z, A_p z$ gives

$$\det(z, A_p z) = u(au + bv/2) - v(-bu/2 - cv) \quad (2.12)$$

$$= au^2 + buv + cv^2 = p(u, v). \quad (2.13)$$

The determinant vanishes precisely when z and $A_p z$ are linearly dependent, which is the projective eigenline condition. $\square$

Remark 2.2 (Why the chart matters). *On $u \neq 0$, Eq. (2.10) reads $u^2 q(\xi) = 0$. At infinity, $[u : v] = [0 : 1]$, one must instead use Eq. (2.5). Replacing $\xi = v/u$ by u/v without transforming the vector-field coefficient reverses the chart convention and can exchange which coefficient detects the point at infinity.*

3 The exact discriminant–Killing identities

3.1 The square of the projective matrix

Define the binary-quadratic discriminant

$$\Delta(p) := b^2 - 4ac. \quad (3.1)$$

Direct multiplication, with no spectral assumption, gives

$$A_p^2 = \begin{pmatrix} b^2/4 - ac & bc/2 - bc/2 \\ -ab/2 + ab/2 & b^2/4 - ac \end{pmatrix} \quad (3.2)$$

$$= \boxed{\frac{\Delta(p)}{4} I_2}. \quad (3.3)$$

Equivalently, Cayley–Hamilton for a traceless two-by-two matrix gives $A_p^2 = -\det(A_p)I_2$, and

$$\det(A_p) = -\frac{\Delta(p)}{4}. \quad (3.4)$$

3.2 Killing normalization from the adjoint trace

The factor two in AP-E2 is fixed by the canonical Killing form, not by a rescaling chosen after the fact.

Lemma 3.1 (Fundamental-trace formula). *For $A, B \in \mathfrak{sl}_2(\mathbb{C})$,*

$$B_{\text{Kill}}(A, B) := \text{Tr}(\text{ad}_A \text{ad}_B) = 4 \text{Tr}_2(AB). \quad (3.5)$$

Proof. Write

$$A = \begin{pmatrix} h & e \\ f & -h \end{pmatrix}, \quad B = \begin{pmatrix} h' & e' \\ f' & -h' \end{pmatrix}. \quad (3.6)$$

Computing the adjoint maps on any fixed (E, H, F) basis gives

$$\text{Tr}(\text{ad}_A \text{ad}_B) = 8hh' + 4(e f' + f e'). \quad (3.7)$$

On the other hand,

$$4 \text{Tr}_2(AB) = 4\{2hh' + e f' + f e'\}, \quad (3.8)$$

which is identical. This fixes the normalization of Eq. (3.5). $\square$

Combining Eqs. (3.3) and (3.5),

$$\boxed{B_{\text{Kill}}(A_p, A_p) = 4 \text{Tr}_2(A_p^2) = 2\Delta(p).} \quad (3.9)$$

Writing $B = \Delta$ would therefore be a normalization error, not a harmless choice of notation.

3.3 Second transvectant and polarization

With the derivative convention used in Route E, the second transvectant of a quadratic with itself is

$$(p, p)_2 := p_{uu}p_{vv} - 2p_{uv}p_{uv} + p_{vv}p_{uu} \quad (3.10)$$

$$= 8ac - 2b^2 = -2\Delta(p). \quad (3.11)$$

Thus

$$\boxed{(p, p)_2 = -B_{\text{Kill}}(A_p, A_p).} \quad (3.12)$$

The bilinear statement is stronger than the diagonal one. Let

$$r(u, v) = a'u^2 + b'uv + c'v^2 \quad (3.13)$$

and polarize the discriminant:

$$D(p, r) := bb' - 2(ac' + a'c), \quad D(p, p) = \Delta(p). \quad (3.14)$$

Then direct multiplication gives

$$\boxed{B_{\text{Kill}}(A_p, A_r) = 2D(p, r), \quad (p, r)_2 = -2D(p, r) = -B_{\text{Kill}}(A_p, A_r).} \quad (3.15)$$

This polarization prevents a false proof that happens to reproduce only a single diagonal example.

4 Zeros, eigenlines, and the null boundary

4.1 Regular and degenerate projective divisors

Over $\mathbb{C}$, a nonzero homogeneous quadratic has a degree-two zero divisor, counted with multiplicity. If $c \neq 0$, its finite affine roots are

$$\xi_{\pm} = \frac{-b \pm \sqrt{\Delta}}{2c} \quad (c \neq 0), \quad (4.1)$$

When $c = 0$ and $b \neq 0$, one root is finite and the other is $[0 : 1]$; when $c = b = 0$ and $a \neq 0$, $[0 : 1]$ is a double root. The homogeneous formulation handles both cases without a singular formula.

Equation (3.3) proves the corresponding Lie-algebra classification:

- if $\Delta \neq 0$, the eigenvalues of A_p are $\pm\sqrt{\Delta}/2$, so A_p is regular semisimple;
- if $\Delta = 0$ and $p \neq 0$, then $A_p \neq 0$ and $A_p^2 = 0$, so A_p is rank-one nilpotent;
- if $p = 0$, then $A_p = 0$, which is neither a nonzero nilpotent orbit nor a degree-two effective zero divisor.

Theorem 4.1 (AP-E2 discriminant/contact equivalence). *For a nonzero $X_p \in H^0(\mathbb{CP}^1, T_{\mathbb{CP}^1})$, the following are equivalent:*

1. *the zero divisor of X_p is the sum of two distinct projective points;*
2. $\Delta(p) \neq 0$;
3. A_p *is regular semisimple;*
4. $B_{\text{Kill}}(A_p, A_p) \neq 0$;
5. *the Route-E contact norm is nonzero.*

The complement inside the nonzero section space is the nilpotent/contact-null cone. Each of its points has one double projective zero. The origin is an additional zero-section stratum and is excluded from both orbit statements.

Proof. The roots of a nonzero binary quadratic are distinct exactly when $\Delta \neq 0$. Equations (3.3) and (3.4) show that $\Delta \neq 0$ is equivalent to two distinct eigenvalues and hence to regular semisimplicity. Equation (3.9) equates this with nonzero Killing norm. Section 6 below proves that the Route-E contact is the same bilinear invariant with fixed nonzero normalization. If $\Delta = 0$ and $p \neq 0$, the preceding rank-one nilpotent argument applies. A quadratic with zero discriminant is a square of a nonzero linear form, so its unique projective root has multiplicity two. Finally, $p = 0$ gives $A_p = 0$ and must be separated before the nonzero orbit classification is stated. $\square$

4.2 Finite, zero, and infinite double roots

The differential of the homogeneous section is

$$dp = (2au + bv)du + (bu + 2cv)dv. \quad (4.2)$$

A double projective root is characterized by

$$p(u, v) = 0, \quad 2au + bv = 0, \quad bu + 2cv = 0. \quad (4.3)$$

The following cases explicitly guard both charts.

Section	Double root	Chart interpretation
$(2u - v)^2$	$[1 : 2]$	finite nonzero $\xi = 2$
v^2	$[1 : 0]$	finite zero $\xi = 0$
u^2	$[0 : 1]$	$\eta = 0$, the point at infinity

All three satisfy Eq. (4.3); restricting to one affine quadratic formula would miss the last case.

Remark 4.2 (Nondegenerate form versus null vector). *The contact matrix is invertible, yet a nonzero p can obey $B(p, p) = 0$. This is ordinary isotropy of a complex symmetric bilinear form, not degeneracy of the form. Degeneracy would mean that some nonzero vector pairs to zero with every vector.*

5 Exact $SL(2)$ covariance

Let

$$g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in SL(2, \mathbb{C}), \quad \alpha\delta - \beta\gamma = 1, \quad (5.1)$$

and define the right substitution

$$p^g(z) := p(gz). \quad (5.2)$$

Expansion gives

$$a^g = a\alpha^2 + b\alpha\gamma + c\gamma^2, \quad (5.3)$$

$$b^g = 2a\alpha\beta + b(\alpha\delta + \beta\gamma) + 2c\gamma\delta, \quad (5.4)$$

$$c^g = a\beta^2 + b\beta\delta + c\delta^2. \quad (5.5)$$

Proposition 5.1 (Conjugacy and invariant discriminant). *Under Eq. (5.2),*

$$A_{p^g} = g^{-1} A_p g, \quad \Delta(p^g) = \Delta(p). \quad (5.6)$$

For $g \in GL(2, \mathbb{C})$, the second equality generalizes to $\Delta(p^g) = \det(g)^2 \Delta(p)$.

Proof. Insert Eq. (5.5) into Eq. (2.9); matrix multiplication gives the first identity. Conjugacy preserves $\det A_p = -\Delta(p)/4$, proving the $SL(2)$ statement. For a general g , either direct expansion or factoring g into a scalar and an $SL(2)$ matrix gives the determinant weight two. $\square$

Scaling the section itself by $\lambda \in \mathbb{C}$ gives the separate covariance law

$$A_{\lambda p} = \lambda A_p, \quad \Delta(\lambda p) = \lambda^2 \Delta(p), \quad B(A_{\lambda p}, A_{\lambda p}) = \lambda^2 B(A_p, A_p). \quad (5.7)$$

Thus the vanishing locus is projectively well defined even though the quadratic invariant has weight two.

5.1 Uniqueness of the invariant contact line

The contact is not one arbitrary matrix among many. In the unnormalized coefficient vector

$$\mathbf{c} = (a, b, c)^T, \quad (5.8)$$

the infinitesimal $\mathfrak{sl}_2$ action can be represented by

$$\rho(H) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix}, \quad \rho(E) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}, \quad \rho(F) = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}. \quad (5.9)$$

Let $M = M^T$ be a general six-parameter symmetric matrix. Invariance means

$$\rho(X)^T M + M \rho(X) = 0, \quad X = H, E, F. \quad (5.10)$$

The H equation permits only $M_{13} = M_{31}$ and M_{22} . The E equation then fixes $M_{13} = -2M_{22}$; the F equation adds no independent freedom. Hence the solution space is one-dimensional:

$$M = \kappa \begin{pmatrix} 0 & 0 & -2 \\ 0 & 1 & 0 \\ -2 & 0 & 0 \end{pmatrix}. \quad (5.11)$$

At $\kappa = 1$, $\mathbf{c}^T M \mathbf{c} = b^2 - 4ac = \Delta$. Equivalently, the exact linear system has rank five in a six-dimensional space of symmetric matrices. This is the first-principles uniqueness of the invariant symmetric contact line.

6 Route-E basis ledger and the factor of two

6.1 Spherical coordinates

Route E uses

$$\mathbf{x} = (x_+, x_0, x_-)^T, \quad K_{\text{tr}} = \frac{1}{\sqrt{3}} \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}. \quad (6.1)$$

The precise map to Eq. (2.8) is

$$a = \frac{x_-}{\sqrt{2}}, \quad b = -x_0, \quad c = \frac{x_+}{\sqrt{2}}, \quad (6.2)$$

or

$$p_{\mathbf{x}}(u, v) = \frac{x_-}{\sqrt{2}} u^2 - x_0 uv + \frac{x_+}{\sqrt{2}} v^2. \quad (6.3)$$

It follows that

$$\Delta(p_{\mathbf{x}}) = x_0^2 - 2x_+x_-, \quad (6.4)$$

$$\mathbf{x}^T K_{\text{tr}} \mathbf{x} = \frac{1}{\sqrt{3}} (x_0^2 - 2x_+x_-). \quad (6.5)$$

Combining this with Eq. (3.9) yields the frozen Route-E identity

$$\boxed{B_{\text{Kill}}(A_p, A_p) = 2\Delta(p) = 2\sqrt{3} \mathbf{x}^T K_{\text{tr}} \mathbf{x}.} \quad (6.6)$$

The matrix is symmetric and invertible:

$$K_{\text{tr}}^2 = \frac{1}{3} I_3, \quad K_{\text{tr}}^{-1} = 3K_{\text{tr}}, \quad \det K_{\text{tr}} = -\frac{1}{3\sqrt{3}}. \quad (6.7)$$

Nevertheless,

$$\mathbf{x} = (1, 1, 1/2)^T \implies \mathbf{x}^T K_{\text{tr}} \mathbf{x} = 0, \quad K_{\text{tr}} \mathbf{x} \neq 0, \quad (6.8)$$

which explicitly distinguishes a nonzero isotropic vector from a degenerate pairing.

6.2 Normalized polynomial coefficients

Now expand the same section in the normalized monomial basis

$$(u^2, \sqrt{2}uv, v^2), \quad p = y_- u^2 + y_0 \sqrt{2}uv + y_+ v^2. \quad (6.9)$$

Thus

$$\mathbf{y} = (y_-, y_0, y_+)^T = \begin{pmatrix} a \\ b/\sqrt{2} \\ c \end{pmatrix} = \frac{1}{\sqrt{2}} \underbrace{\begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{pmatrix}}_R \mathbf{x}. \quad (6.10)$$

Because $R^T K_{\text{tr}} R = K_{\text{tr}}$,

$$\mathbf{y}^T K_{\text{tr}} \mathbf{y} = \frac{1}{2} \mathbf{x}^T K_{\text{tr}} \mathbf{x}. \quad (6.11)$$

Therefore the same Killing invariant has two correct coefficient formulas:

$$\boxed{B = 2\sqrt{3} \mathbf{x}^T K_{\text{tr}} \mathbf{x} = 4\sqrt{3} \mathbf{y}^T K_{\text{tr}} \mathbf{y}.} \quad (6.12)$$

Coordinates	Section convention	Killing formula
$\mathbf{c} = (a, b, c)$	$au^2 + buv + cv^2$	$B = 2(b^2 - 4ac)$
$\mathbf{y} = (a, b/\sqrt{2}, c)$	$y_-u^2 + y_0\sqrt{2}uv + y_+v^2$	$B = 4\sqrt{3}\mathbf{y}^T K_{\text{tr}}\mathbf{y}$
$\mathbf{x} = (x_+, x_0, x_-)$	Eq. (6.3)	$B = 2\sqrt{3}\mathbf{x}^T K_{\text{tr}}\mathbf{x}$

Using the spherical coefficient $2\sqrt{3}$ on $\mathbf{y}$ loses a factor of two. Omitting the $\sqrt{2}$ in Eq. (6.2) changes $\Delta = x_0^2 - 2x_+x_-$ into the wrong expression $x_0^2 - 4x_+x_-$. Both errors are explicit negative controls in the regression suite.

6.3 Complex symmetric is not Hermitian

The Route-E contact is

$$K_{\text{tr}}(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T K_{\text{tr}}\mathbf{y}, \quad (6.13)$$

which is complex bilinear and symmetric. Replacing it by $\mathbf{x}^\dagger K_{\text{tr}}\mathbf{y}$ introduces complex conjugation and destroys the holomorphic invariant polynomial. In particular,

$$(\lambda\mathbf{x})^T K_{\text{tr}}(\lambda\mathbf{x}) = \lambda^2 \mathbf{x}^T K_{\text{tr}}\mathbf{x}, \quad (6.14)$$

whereas the Hermitian expression scales as $|\lambda|^2$. These are different representations and cannot be exchanged on complex samples.

7 The H3 versus H3⁺ boundary

An invariant nondegenerate symmetric bilinear form does not by itself imply a nonzero Killing form. Let $\mathfrak{a} = \mathbb{C}e$ be a one-dimensional abelian Lie algebra. Then

$$[e, e] = 0, \quad \text{ad}_e = 0, \quad B_{\text{Kill}}(e, e) = \text{Tr}(\text{ad}_e^2) = 0. \quad (7.1)$$

However, the form $C(e, e) = 1$ is symmetric, invariant, and nondegenerate, because every bracket vanishes. Hence the original H3 condition “there is a nondegenerate invariant symmetric pairing” admits this genus-one/abelian counterexample. Exact selection of the three-dimensional $\mathfrak{sl}_2(\mathbb{C})$ carrier requires the stronger H3⁺ condition that the contact is the nondegenerate adjoint-trace/Killing line.

Corollary 7.1 (Precise AP-E2 scope). *Theorem 4.1 proves an exact equivalence on the H3⁺-selected $\mathbb{CP}^1$ branch. It neither derives H3⁺ from H3 nor rules out the abelian H3 branch.*

8 The 30-check fail-closed regression

8.1 Reproducibility contract

The executable card is

```
route_f/code/verify_ap_e2_discriminant_regression.py.
```

From the repository root, run

```
python3 route_f/code/verify_ap_e2_discriminant_regression.py.
```

It writes

route_f/output/ap_e2_discriminant_regression.json
route_f/output/ap_e2_discriminant_regression.md.

The seed is 20260714. Exact tests use `fractions.Fraction`; irrational and complex tests use `mpmath1.4.1` at 100 decimal places. The program writes its complete report before returning exit status one if any check fails.

8.2 Coverage by group

Group	Checks	Fail-closed content
A0	4	Critical source hashes, invariant-card stable digest, Audit-0 builder binding, and frozen theorem/proof-block hash.
A1	10	Exact rational A^2 , Killing, transvectant, polarized contact, one-dimensional invariant line, $\det(z, Az) = p(z)$, $SL(2, \mathbb{Z})$ invariance and conjugacy, and section-scaling weight.
A2	5	Finite and infinite double roots, real/complex/asymmetric regular roots, eigenline agreement, vanishing first jet, zero-section guard, and a 10^{-70} near-null perturbation.
A3	4	Route-E K_{tr} normalization, inverse and isotropic-vector guard, spherical/normalized-basis factor two, and full H, E, F invariance.
A4	1	256 seeded complex samples evaluated at 100 decimal places.
A5	4	Required rejection of missing $\sqrt{2}$, missing Killing factor two, wrong normalized-basis coefficient, and Hermitian substitution.
A6	2	The abelian H3 counterexample and the immutable non-promotion Boolean.
Total	30	Every check must pass; otherwise the process exits nonzero.

The exact randomized layer uses 256 seeded triples in $\mathbb{Q}^3$, 256 seeded pairs for the polarized identity, 256 exact projective fixed-point tests, and 256 determinant-one integer substitutions. Since all operations are in $\mathbb{Q}$, “zero residual” means an exact equality, not a floating tolerance.

The complex layer separately samples 256 points in $\mathbb{C}^3$. Its largest intrinsic normalized residual is

$$1.67930436345868921 \times 10^{-101}. \quad (8.1)$$

The Route-E JSON card stores ordinary double-precision decimals, so its separate maximum source-card residual is

$$6.14719575160384581 \times 10^{-17}, \quad (8.2)$$

well below the declared 5×10^{-15} source tolerance. Keeping these two residuals separate prevents the source serialization precision from masquerading as a limitation of the 100-digit identity test.

For the near-null probe

$$p_\varepsilon = u^2 + 2uv + (1 + \varepsilon)v^2, \quad \varepsilon = 10^{-70}, \quad (8.3)$$

the audit resolves

$$\Delta(p_\varepsilon) = -4 \times 10^{-70} \quad (8.4)$$

and obtains zero recorded matrix-identity residual at the working precision. This tests the boundary from the regular side instead of evaluating only the exact cone.

8.3 Negative controls

A regression is incomplete if it tests only the intended formula. AP-E2 therefore demands that four intentionally wrong conventions disagree on a fixed generic complex point.

Wrong convention	Expected diagnosis	Residual
$a = x_-, b = -x_0, c = x_+$	missing $\sqrt{2}$	1.0×10^1
$B = \Delta$	missing Killing factor two	1.2×10^1
$B = 2\sqrt{3} \mathbf{y}^T K_{\text{tr}} \mathbf{y}$	spherical factor used on normalized coefficients	1.2×10^1
$\mathbf{x}^\dagger K_{\text{tr}} \mathbf{x}$	Hermitian form substituted for bilinear contact	1.56205×10^1

Detection of each wrong formula is a pass. Accidental agreement is a failure. An additional injected bad theorem hash was tested during development; it produced 29/30 and a nonzero process exit before the valid artifact was regenerated.

8.4 Current result and provenance lock

The current card reports

$$\boxed{30/30 \text{ checks pass.}} \quad (8.5)$$

Its non-promotion field remains

$$\text{physics_promotion_allowed=false.}$$

The frozen theorem/proof block has SHA-256

$$3999b88f3fe7e396948c9695939ec5826d74cfc6d06f6b26ed05d3b0e0c4d094.$$

The audit also recomputes the stable digest of the Route-E invariant card and checks that the card-recorded Audit-0 builder hash equals the current builder. Thus a green result requires both correct arithmetic and an intact convention source chain.

9 Claim ledger and handoff to AP-E3

Claim	AP-E2 status	Evidence or missing input
Global vector fields on $\mathbb{CP}^1$ are binary quadratics	proved	Two-chart regularity, Eqs. (2.3)–(2.4).
Zeros of p equal eigenlines of A_p	proved	Determinant identity (2.10).
Two distinct zeros, regular semisimplicity, and non-null contact are equivalent	proved on carrier	Theorem 4.1.
The null cone consists of nonzero double-zero/nilpotent sections	proved	Matrix square and first-jet tests; zero section excluded.
The Route-E factor is $B = 2\sqrt{3} \mathbf{x}^T K_{\text{tr}} \mathbf{x}$	proved	Killing trace plus spherical map.

Claim	AP-E2 status	Evidence or missing input
Normalized coefficients use $B = 4\sqrt{3}y^T K_{\text{tr}}y$	proved	Exact basis map (6.10).
H3 alone selects the three-dimensional carrier	refuted	One-dimensional abelian invariant-pairing counterexample.
H3 ⁺ has a microscopic dynamical origin	open	Requires a correlator, messenger, anomaly, or UV derivation.
The two divisor points are physical particles, signs of energy, or families	not derived	No action or observable map supplies this identification.
The projective sector has level magnitude $ k = 2$	candidate magnitude derived; sign/UV open	The AP-E3 Mott–Hund pair gives a ket line $\mathcal{O}(-2)$ and dual prequantum line $\mathcal{O}(2)$; the signed orientation and exactly-two UV rule remain absent.

AP-E2 is therefore mathematically closed at the regression level. AP-E3 should not repeat this discriminant calculation. It should instead test several genuinely distinct microscopic routes to level two:

1. derive a charge-two collective coordinate and its Berry one-form from a fully normalized microscopic action;
2. integrate out explicitly specified gapped fields and compute the induced worldline Chern–Simons/Berry level, including anomaly and sign checks;
3. derive a pair-selection rule that leaves a level-two composite sector while keeping the macroscopic Q-ball charge separate;
4. reject every proposal for which the induced level is not integral, is shifted by an uncanceled anomaly, or has a Landau-level gap comparable to the portal or background scale.

None of these AP-E3 mechanisms can be inferred from $\Delta = B/2$ alone. That separation is the principal scientific output of AP-E2.

10 Conclusion

Starting only from the two charts of $\mathbb{CP}^1$, AP-E2 derives the complete chain

$$\det(z, A_p z) = p(z), \quad A_p^2 = \frac{\Delta}{4}I, \quad B_{\text{Kill}} = 2\Delta, \quad (p, p)_2 = -2\Delta. \quad (10.1)$$

It proves that the discriminant, projective two-center divisor, regular/nilpotent stratification, Killing norm, and Route-E contact are one invariant statement on the conditional $\mathbb{CP}^1$ carrier. The chart at infinity, zero-section exception, complex-bilinear character, and spherical/normalized-basis factor two are all explicit rather than implicit.

The 30-check regression now freezes these conclusions against known convention failures and source drift. This completes AP-E2 without promoting the bridge. AP-E3 now supplies a conditional Mott–Hund pair with $|k| = 2$; its displayed aligned action has $k = -2$, while the desired sign, the physical origin of H3⁺, and the UV principle selecting exactly two constituents remain logically separate open problems.